1. Purpose:

To specify the required production process for breathing circuit and its accessories to ensure the good quality product during production & before releasing to the market.

2. Scope:

This procedure is applicable for production process of breathing circuit and its accessories producing by Saudi mais company.

3. Definition:

- **Breathing tube** – non-rigid / extendable or corrugated or smooth bore tube used to convey gases or some ventilators.
- **Adaptor** – intended to be inserted into the end of a breathing tube having a conical connector Ø 22 mm, Ø 15 mm
- **Plain end** – is designed to fit directly over a male conical connector dia. Ø 22 mm.
- **Patient end** – is intended to be connected to the Y-piece, appropriate component near the patient
- **Machine end** – is intended to be connected to the anesthetic workstation, ventilation or other breathing attachment furthest from the patient

4. Breathing Circuit Contains accessories:

- **Breathing Filter** is intended use to stop bacteria and viruses being transferred to and from the patient and preventing cross infection.
- **Catheter Mount** is connected between the patient and the breathing system
- **Swivel Connector** used to insert a flexible fiberscope.
- **MDI Adaptor (**metered dose inhaler) used to deliver medication to ventilated patients
- **Breathing Bag** to allow respiratory monitoring and/or assistance.

5. Responsibilities:

5.1 Quality controller to sure that all quality points are met by specification & required.

5.2 Machine operators are responsible to produce as per required quantity and with good quality.

5.3 Machine operators and assembly peoples are responsible to produced good quantity and good quality products.

6. Production Procedure:

6.1 Receiving Raw Material:

- Ensure the receiving slip from supplier is correct materials in order to specification
- Inspect the materials condition, must be in good condition, no damaged at all Bags with material.

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- Inspect the materials which written on receiving slip from supplier.

6.1.1 Inspection & test report procedure for Raw Material:

Work Description	Requirements	Method of Check
Clarity / Appearance	Clear / Good Appearance	Visual
Condition of package	Good package without damage	Visual
Free from Irregularities	No flashing / No short shot / No back sport	Visual
Appearance of Granules	Good / Transparent / Colored	Visual & Physical
Granules formation	No lobes formation	Visual
Lot number printed status	Clear printed	Visual
Arrangement of materials bag	Should be specified clear label	Visual

7. Injection Machine Procedure

- Production controller is responsible to issue required raw material for breathing adaptor, y connector, end cap, breathing filter housing, swivel connector adaptor and MDI Adaptor.
- Production controller is responsible to make sure that the received material is same as per issue slip.
- Production controller is responsible to issue production order for product parts for berating tube and accessories.
- Maintenance department is responsible to set the required mold on specified machine and set the standard parameters for each component.
- Quality controller is responsible to verify the parameters and make sure that the required standard parameters are set.
- Machine operator is responsible to fill the material into machine hopper by using vacuum pump and fill hoppers with material up to the required capacity.
- Machine operator is responsible to give initial sample from each cavity of the mold to quality controller for inspection and get approval from quality controller.
- Quality controller is responsible to inspect the product against specification and make sure that the products is meeting the standard specification.
- The Quality Controller must collect the samples as per SOP 8-2-6 based on AQL.

7.1.1 Breathing Adaptor:

- Quality controller is responsible to make sure that the breathing adaptor must be free from flashing, bends, no color faded, no black spot with the breathing tube, accurate in dimension.

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7.1.2 Y connectors:

- Must be transparent no black spot, free from flashing, no cracks, no damage, and good fitting with breathing tubes & accurate in dimension no pull-out form berating adaptor.

7.1.3 End plug:

Must be good fitting with female port no leakage after fitting free from crack and damage.

7.1.4 Breathing filter housing and cover:

Clear and transparent, no crack, no damage, no black spot, free from flashing and dust particle, round surface, good fitting with housing and cover, accurate in dimension.

7.1.5 T-connector, adaptor and cap for Swivel:

Must be clear and transparent, smooth surface, good fitting, no loose fitting, no cracks, no damage, no Black spot, no flashing, accurate in dimension.

7.1.6 Adaptor for catheter mount:

Smooth surface, no damage, good fitting, no cracks, no flashing, accurate in dimension.

7.1.7 MDI Adaptor:

Clear and transparent adaptor, accurate in dimension OD & ID, no crack, no damage, without any irregularity's defects.

7.1.8 Male and Female connector:

Quality controller is responsible to make sure that male connector should be clear and transparent free from crack and damaged with clear thread marks and without any irregularity's defects & Female connector should be clear and transparent good fitting with male connector strong fitting no leakage and free from damaged and crack.

8. Extrusion Machine procedures:

8.1 <u>Breathing tube, Corrugated Tube, Corrugated Extendable and CO2 Sampling line:</u>

- Machine operator is responsible to fill the machine hopper with material which has issued and make sure that the material is same as per production order.
- Maintenance coordinators are responsible to set the required die for each type of breathing tubes as per customer requirement and set the standard parameters.
- Machine operator is responsible to clean the machine surface and make sure each station of machine should be clean and dust free.
- Machine operator is responsible to submit initial sample to quality controller for inspection.
- Quality controller is responsible to inspect the breathing tubes against specification and make sure that the breathing tubes are meeting the standard specification.

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- The Quality Controller must collect the samples as per SOP 8-2-6 based on AQL.
- The quality controller is responsible to check the quality of the products and make sure that tube with in accurate in dimension length, ID, OD, and length, and tube should be no black spots, no fish eyes, no broken, accurate distance between the ribs and free from irregularities defects.
- Quality controller is responsible to monitor the production every 2 hours and record the results in product in process control form.
- Machine operator is responsible to arrange the good products in clean transparent polyethylene bag.
- Quality controller is responsible to make inspection tag and place inspection tag on the bag with approval stamp as "QC APPROVED".
- Production controller is responsible to arrange the good products in specified rack or transfer to mais components storage facility.

8.2 Receiving Bulk components:

- Ensure the receiving slip from supplier is correct components in order to specification**.**
- Inspect the carton condition, must be in good condition, no damaged at all cartons with products**.**
- Inspect the product condition it must be free from dust particle, no damaged, no breakage, no leakage, with all components.
- Take samples & inspect in order to AQL procedure **SOP 8-2-6**, components description, specification with apply the AQL sampling plans for normal inspection as per inspection report and record into OBM inspection report.

8.2.1 Breathing Bag

- Quality inspector is responsible to make sure that the breathing bag should be no damage, no holes, no leakage and free from dust particle, and should be accurate in capacity.

8.2.2 Paper filter and HME filter

- Filter should be accurate in dimension, round cutting edges, good fitting in filter housing, no damage and free from irregularities defects.

8.2.3 O –ring (Black)

- Silicon black o ring round shape without damage, no broken, accurate in dimension, good fitting in swivel adaptor.

9 Assembly process:

9.1 Assembly of Breathing tube with adaptor

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- Production controller is responsible to issue breathing tube, and adaptor and responsible to issue production order for assembly.
- Fill the glue (cyclohexane) into small petri dish with sponge and make sure glue should be with valid expiry date.
- Then dip the tube into the glue and connect immediately to circuit adaptor and make sure that no excess gluing in both sides used and tube should be touch internal wall of adaptor.
- After gluing, insert the Y connector into adaptor with circuit tubes and winding it by two lobes formed.
- Put glue to other end of tube and insert the adaptor for both tube and make sure adaptor is tight fit with tube not pull out.
- Fix end plug with female port and make sure it should be tight fit.
- Quality controller is responsible to check the products

Quality controller is responsible to record the results in in process control form.

- Production controller is responsible to arrange the assembled products in clean transparent polyethylene bag.
- Quality controller is responsible to make inspection tag and place inspection tag on the bag with approval stamp as "**QC APPROVED**".
- Production controller is responsible to arrange the good products in specified rack.

9.1.1 QC In Process Control for Breathing Circuit:

Description / Components	Requirements	Method of Check
Corrugated tube, smooth bore	Good appearance, soft tubing	Visual
If bending tube	Flexible, No damage	Visual & Physical
Tube over ribs attachment	Strong, ribs should not come out,	Cometech Machine
Tube ribs quality	Smooth, w/o roughness	Visual & Physical
Tube length	160, 180, 200 cm	Scale
Tube ID X OD (mm)	Adult(22x25), Pedi(15x19), Neo(10x15)	….
Tube Quality	W/o holes, damage, sink mark	Visual & Physical
Elbow connector with End cap	Good fitting w/o cracks, holes	Visual & Physical
Fitting elbow + Y-connector	Good fitting, No loose	Visual & Physical
Gluing with tube + adaptor	Strong w/o pull out, by force of Min 45±2N	Cometech Machine
Y-Connector part + caps	Proper fitting w/o damage	Physical
Adaptor fitting with Y-connector	Tight fit, No loose	Cometech Machine

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Tube sheet Thickness	0.55mm - 0.70mm	Caliper
Ribs Distance	5.8 mm - 6.2mm	Caliper
Ribs Thickness	2.3 mm -2.5mm	Caliper
Tube Leak test	Required pressure 6KPA for 5 mints	Leak test Machine

9.2 Assembly for Catheter mount and swivel connector

- Production controller is responsible to issue parts and production order for assembly.
- Fill the glue (cyclohexane) into small Petri dish.
- Put silicon O-ring (black) in swivel adaptor and make sure it's properly fixed in ring inside the adaptor.
- Press the swivel adaptor in T-piece connector to assemble both parts and put the cap across over the adaptor to close the T-piece.
- Dip the corrugated tube extendable in glue and fix with T-piece connector and other end fixed with corrugated adaptor.
- After assembly make sure that no excess glue is in the surface of the tube and adaptor.
- Quality controller is responsible to check the products randomly as per in process control form assembly and ensure that products is good quality and meets the specification criteria.
- Production controller is responsible to arrange the assembled products in clean transparent polyethylene bag.
- Quality controller is responsible to make inspection tag and place inspection tag on the bag with approval stamp as "**QC APPROVED**".

Production controller is responsible to arrange the good products in specified rack.

9.2.1 QC in Process Control for Catheter Mount:

Description / Components	Requirements	Method of Check
Clarity / Appearance	Clear / Good Appearance	Visual
Free from Irregularities	No short / No flashing	Visual & Physical
Tube Condition	Smoothness / Flexible	Visual & Physical
Tube Length	Should be 15 cm	Scale
Tube Connector ID x OD	Should be (ID 15 x OD 22 mm)	Caliper
Connector ID x OD	Should be (ID 15 x OD 22 mm)	Caliper
Mount direction into Conn.	Correct direction, good fit into connector	Visual & Physical
Gluing status	Strong glue / Not pulled out	Universal Test Machine
Condition of Products	Good & Flexible movement	Visual & Physical

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9.2.2 QC In Process Control for Swivel connector

Description / Components	Requirements	Method of Check
Clarity / Appearance	Clear / Good Appearance	Visual
Swivel Condition	No crack, No damaged, smooth surface, No defected	Visual & Physical
Free from Irregularities defect	No flashing, No short shot, No black spot, No white spot	Visual & Physical
Swivel Type	Disposable / Reusable	Visual
Connector Diameter / Status	Dia. 22 mm connector, good fitting for Endotracheal adaptor	Caliper
Suction connector status	Dia. 15 mm connector, good fitting for adaptors I.D. O.D.	Caliper
Connector rotation movement status	Both should smooth rotation, No sharp edge or flashing	Visual & Physical
Connector with cap status	Elastomeric cap with good fitting for connector, OD & ID perfect Dia.	Visual & Physical & Caliper
Condition of products	Must be good condition at all parts	Visual & Physical

9.3 Breathing Filter Assembly

- Production controller is responsible to issue breathing filter parts and filter and also responsible to issue production order for assembly.
- Take breathing filter housing lower part and make sure housing is free from crack damage and free from dust particle.
- Fix paper filter and HME filter in luer part housing and make sure its properly fit inside the housing.
- Close housing with upper part and make sure is properly fix with each other.
- Pout port cap on female port and make sure its not leakage and tight fit.
- Then Seal breathing filter with ultrasonic sealing machine by using standard parameters as per given table and make sure filter properly sealed without any leakage.

Welding Time	Delay Time	Hold Time	Pressure
0.5-1.5 Sec	0.5-1.5 Sec	0.5-1.5 Sec	5.5-6.5Kg/cm^2

- Quality controller is responsible to record the results in in process control form.
- Production controller is responsible to arrange the assembled products in clean transparent polyethylene bag.
- Quality controller is responsible to make inspection tag and place inspection tag on the bag with approval stamp as "**QC APPROVED**".
- Production controller is responsible to arrange the good products in specified rack.

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9.3.1 In process control for Breathing filter

Description / Components	Requirements	Method of Check
Clarity / Appearance	Clear / White Color	Visual
Housing Condition	No crack, No damage	Visual
Lower Housing Condition	No crack, No damage with goo fitting	Visual
Upper Housing Condition (cap)	No crack, No damage with good fitting	Visual
Lock Condition / status	Tight fitting, No leakage	Visual & Physical
Bacterial filter for adult	Non-woven filter Dia- 59±1mm	Physical & Caliper
Humidifying paper for Adult	Dia 53 ±1mm / Width 15mm	Caliper
Bacterial filter for pediatric	Non-woven filter Dia- 44±1mm	Physical & Caliper
Humidifying paper for pediatric	Dia 30 ±1mm / Width 15mm	Caliper
Filter position / fitting	Good position good fitting	Visual & Physical
Deformity of housing on filter	No defect at all	Visual & Physical
Fitting of filter and paper inside of housing	Correct position	Visual & Physical
Status of humidifying paper	No movement	Visual & Physical
Top filter condition	No movement after fixing	Visual & Physical
Capping status	Tight fitting	Visual & Physical
End plug status	Good fitting, good twisting	Visual & Physical
Condition of filter after fixing	Good position, No movement	Visual & Physical

9.4 Assembly for CO2 sampling Line

- Take Co2 sampling line tube and connectors and make sure connector should be free from crack damage and free from dust particle.
- Dip tube in glue and insert the tube in male connector and another end with female.
- Keep for 1 minutes for dry and then place in bag.
- Quality controller is responsible for tensile test to make sure that the gluing is strong and meets the standard requirements and record the results in in process form.
- Quality controller is responsible to put inspection tag as approved.

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- Production controller is responsible to keep in designated area.

9.4.1 Q.C In-process control for CO2 Sampling Line

Appearance / Clarity	Transparent / Kink Resistance	Visual
Dia. OD x ID mm	OD 3.5mm X ID 1.5mm	By measurement
Length requirement	3 Meters	By measurement
Fitting status of Male / Female connector	Good fitting	Physical
Gluing status of both connector	Strong / No pulled out, good fitting in both sides	Tensile
Leakage	No leakage	Visual
Packaging condition	Secure and safety	Visual

10. Leak Test Procedure:

10.1 Breathing tube leak test Smooth Bore & Corrugated or Extendable corrugated

- Machine operator is responsible to make sure that berating tube smooth bore components are properly assembled.
- Breathing tube leak test performed as per standard requirements BS EN 5367- ANNEX-E.
- Connect the tube with calibrated leak tester device adaptor and make sure that the tube adopter is properly fixed with leak tester device adaptor.
- Parameters Setting on Leak Tester Machine **Model: 3200** for Breathing Circuit All types as per WI-8-2-6-01.

Pressure Setting For Breathing Circuit	Charge Time (±2 sec)	Test Duration Time (± 2sec)
18 - 22 KPA	15 **sec**.	10 **sec**.
180 - 220 mbar	15 **sec**.	10 **sec**.
2.6 – 3.2 psi	15 **sec**.	10 **sec**.

- Parameters Setting on Leak Tester Machine **Delta** for Breathing Circuit All types as per WI-8-2-6-02.

Pressure Setting For Breathing Circuit	Charge Time (±2 Sec)	Test Duration Time (±2 Sec)
30 – 32 KPA	10 sec.	12 sec.
300 – 320mbar	10 sec.	12 sec.
4.35 – 4.64 psi	10 sec.	12 sec.

- Apply pressure and maintain pressure for 5 minutes, and wait until the machine give red and green light indication.

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- If the machine shows red light, then it's not acceptable.
- If the machine shoes green light means acceptable.
- Quality controller is responsible to make inspection tag and place inspection tag on the bag with approval stamp as "**QC APPROVED**".
- Production controller is responsible to arrange the good products in specified rack.

10.2 Breathing Filter Leak Test:

- Quality controller is responsible to arrange equipment for breathing filter leak test.
- Quality controller is responsible to get sample randomly from final finished products.
- Breathing filter leak test performed manually by using 60ml syringe.
- Block breathing filer inlet and outlet port with specified connector and make sure its properly blocked the filter ports.
- Connect tube with syringe and remove end plug form Female port and connect Tube MLL connector with Female connector.
- Dip filter in tap water and apply pressure by syringe for 10 second if any bubble in water during applying pressure the filter is leak and not acceptable.
- Breathing filter should be leakage free.

10.3 Security attachment of Adaptor with Breathing tube:

- Quality controller is responsible to get sample from final finished products for testing of security attachment of adaptor with breathing tube.
- Security attachment of adaptor with breathing tube test performed as per standard requirements BS EN 5367- ANNEX-C & D.
- Apply a tensile load of (45 ± 2) N at a rate of (50 ± 5) mm min-1 and maintain that load for 1 min at a point not less than 150 mm from the end of the tube along the linear axis of the tube, and note whether the tube becomes detached from the adaptor.
- Adaptor should not detach from tube during testing period.
- Quality controller is responsible to make inspection tag and place inspection tag on the bag with approval stamp as "**QC APPROVED**".
- Production controller is responsible to arrange the good products in specified rack.

11. Packaging and Labeling:

11.1 Pouch Packaging (Breathing Circuit Smooth Bore, Corrugated and Extendable Corrugated with breathing bag (*optional*)).

- Production controller is responsible to issue production order for packaging breathing circuits in pouch.

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- Breathing circuit packed in pouch with label and then sealed by manual sealing machine. The label contains the following information's.
  Product name and code no, reference no, size, lot no, type, temperature, logo, production date, expiry date, etc.
- Quality controller is responsible to make sure that the pouch sealed properly and printed information are correct.
- Breathing circuit direct packed in carton.

11.2 Packaging by machine of Catheter mount, Breathing filter, MDI, Swivel Connector:

- Production controller is responsible to issuer production order for packaging machine.
- Machine operator is responsible to set the required die on specified machine and set standard parameters.
- Machine operator is responsible to set the cliches as per approved label as per required in packaging order.
- Machine operator is responsible to give initial sample for quality controller inspection.
- Quality controller is responsible to make sure that package forming sealing, cutting, printing and make sure the printed information is correct as per packaging order and approved cliches.
- packaging should be intact & in good condition.
- Then pack in small boxes and carton.
- The boxes and carton must be with label and detail information.
- Production controller is responsible to arranged the packed products on pallet and transfer to sterilization area.

12 Sterilization:

- The finished products are sterilized as per work instruction WI-7-5-1-1.

13 Final Inspection & Release:

- The Quality controller must perform inspection and testing on the unloaded products Sterilization chamber by using a specific Form for Final Control after sterilization.
- The Quality Controller must use General Inspection for Visual checking and Special Inspection for Physical Checking.
- The Quality Assurance Manager or Quality Control Manager have to review the completed cycle parameters and inspection reports both documents have to be sign.
- After confirmation all the test from the Lab to release the sterilized products, the document once again has to review the quality control manager then sign.

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- The production department responsible to record the quantity of products before transferring to the finished goods ware-house and then it could be released to the customer.

14 References

- **BS EN 5367** Anesthetic and respiratory equipment — Breathing sets and connectors
- **BS EN ISO 9360-1** Anesthetic and respiratory equipment — Heat and moisture exchangers (HMEs) for humidifying respired gases in humans.
- **BS EN ISO 23328-1** Breathing system filters for anesthetic and respiratory use.
- In process control for Breathing Circuit. : F- 823-i8
- In process control for Breathing Filter : F- 823-i47
- In process control for Catheter Mount : F- 823-i39
- In process control for Swivel Connector : F- 823-i49
- In process control for Manual Packaging : F-823-i4-1
- Final Control for Breathing Circuit : F-8-2-3-F27

15. Modification / Documentation change History:

Date	Changes	Revisions
15.03.2018	Updated point 6 and Revision History Table Added	4
10.12.2019	Procedure updated	5
01.01.2021	Performance testing process updated	6
08.12.2022	Updated Section 10. (Leak Testing Parameters) and Section 14. (References)	7
21.10.2025	Procedure updated & Performance testing process updated	8

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